

Unless stated otherwise, all problems are worth 4 points, and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Using the symbolic representations

m : I love math.

c : It is hot in hades.

g : I went to the game.

express the following statements in symbolic form.

a. I went to the game or it is hot in hades.

b. If I love math then it is not hot in hades.

2. Consider the statement, "If Socrates is a man then Socrates is mortal." Use (a), (b), and (c) to complete parts 2a and 2b.

(a) If Socrates is not a man then Socrates is not mortal.

(b) If Socrates is mortal then Socrates is a man.

(c) If Socrates is not mortal then Socrates is not a man.

a. Identify the converse of the statement. _____

b. Identify the contrapositive of the statement. _____

3. For parts 3a and 3b, identify the negation of the provided statement.

a. Some students like football.

b. All pilots can fly a plane.

(a) All students like football.

(a) All pilots can fly a plane.

(b) No students like football.

(b) No pilots can fly a plane.

(c) Some students like football.

(c) Some pilots cannot fly a plane.

(d) Some students do not like football.

(d) Some pilots can fly a plane..

4. [8 points each] Use a truth table to determine if the following arguments are valid.

a. 1. $q \vee r$

2. r

$\therefore \sim q$

		1	2		c	Argument
q	r	$q \vee r$	r	$1 \wedge 2$	$\sim q$	$(1 \wedge 2) \rightarrow c$
T	T					
T	F					
F	T					
F	F					

- b. t : You register early. 1. If you register early, then you will receive a discount.
 d : You receive a discount. 2. You register early.
-
- Therefore, you will receive a discount.

		$\boxed{1}$	$\boxed{2}$		$\boxed{c}$	$\boxed{\text{Argument}}$
r	d			$1 \wedge 2$		$(1 \wedge 2) \rightarrow c$
T	T					
T	F					
F	T					
F	F					

5. Let $U = \{i, l, o, v, e, m, a, t, h\}$ be the universal set, and $A = \{l, o, v, e\}$ and $B = \{v, e, m, a, t, h\}$ be subsets of U . Find the indicated set, and draw a Venn diagram corresponding to the indicated set.

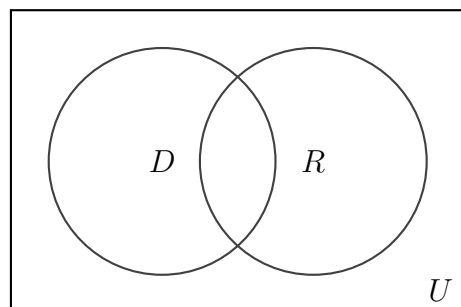
a. $A \cup B$

b. $A \cap B$

c. A'

6. A town has 1000 residents. Of the residents, 600 have voted Democrat, 500 have voted Republican, and 200 have voted for both parties.

- a. Use the Venn diagram, and write the appropriate numbers in each region. b. How many residents have voted Democrat but not Republican?



- c. How many residents have voted Democrat or Republican?

- d. How many residents have voted neither Democrat nor Republican?

7. [2 pts each] Find the value of each of the following:

a. $6!$

c. ${}_5P_3$

b. $\frac{10000!}{9999!}$

d. ${}_5C_3$

12. Two coins are tossed. Find each of the following:

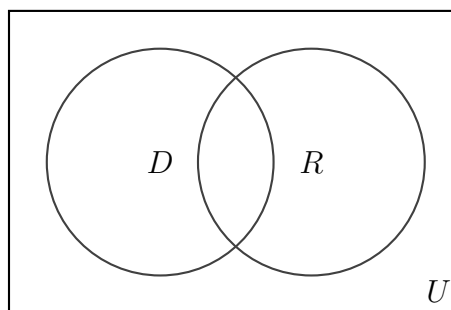
- a. the sample space
- b. [2 points] the event, E_1 , that at least one is tails.
- c. [2 points] the probability of E_1 .

13. A bag of candy has 10 jelly beans. 6 of which are red and 4 of which are green. If you reach into the bag and randomly select 3 jelly beans, find the probability of the following: *Just write the formulas.*

- a. all 3 are green.
- b. exactly 2 are green.

14. Of all the residents in a town 60% have voted Democrat, 50% have voted Republican, and 20% have voted for both parties.

- a. What is the probability that a randomly selected resident has voted Democrat but not Republican?
- b. What is the probability that a randomly selected resident has voted Democrat or Republican?



15. A group of men and women were asked whether they preferred MacGyver or Oprah. Use the results provided in the table to answer the following questions.

	MacGyver	Oprah	NA
Men	40%	8%	2%
Women	10%	35%	5%

- a. Find the probability that a man prefers Oprah.
- b. Find the probability that a person who prefers MacGyver is a women.

16. In Prof. Phails history class, 40% of the students pass and the rest fail. Of the students who fail 55% are male and the rest are female. Of the students who pass 45% of the students are male and the rest are female.

- a.** draw a tree diagram describing the probability **b.** find the probability that a student who fails is female.

- c.** find the probability that a student fails and is female.

17. Use the provided raw data set to complete the following:

1 0 2 2 3 4 1 1

- a.** Find the median. **c.** Find the mode.
- b.** Find the mean. **d.** Find the standard deviation.

18. Create a histogram using the provided frequency distribution table.

hours	freq.
$0 \leq x < 5$	5
$5 \leq x < 10$	10
$10 \leq x < 15$	15
$15 \leq x < 20$	10
$20 \leq x < 25$	2

19. Find the following probabilities,

a. $p(z < -.81)$

b. $p(-.81 < z)$

c. $p(-1.5 < z < -.81)$

20. A population is normally distributed with mean 95 and standard deviation 10. Find the following probabilities.

a. $p(x < 80)$

b. $p(x > 110)$

21. Find c such that each of the following is true.

a. $p(z < c) = .55$

b. $p(z < c) = .3745$

c. $p(z > c) = .3745$

22. An IQ test is normally distributed with mean 100 and standard deviation 15. Find the percentage of the with a score less than 80.